Companion to **The Shape of Gravity** · **Magnetism** · The Shape of the Universe — single home for the global-sector synthesis; the gravity law itself belongs to *The Shape of Gravity* and is cited, not re-derived, here.

The Scalar Field

The global sector, and why it is a gravity object: what "the scalar" actually is in this framework — not a wave, not a fifth force, but the normalization degree of freedom of the local clock-rate law — and the one number that decides whether it is physical.

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> **One-line claim.** "The scalar" is not something that travels fast. It is the *global normalization of the local clock-rate law* — a constraint, not a wave — and it does not sit beside gravity, it sits *inside* gravity, as the scale `E★` in `F = 1 + E/E★`. Whether it is physical is not a question about exotic new physics; it is a one-parameter question, `τ_global`, which this framework can now pose exactly, has measured both endpoints of in-model, and cannot yet decide.

1 · Three answers to one bad question

"Is there a scalar field?" is asked in three incompatible dialects, and most of the confusion in this area is people answering a different one than was asked.

- **The fringe dialect.** A "scalar wave" is a longitudinal, non-transverse carrier that standard electromagnetic bookkeeping drops, and which propagates instantaneously or superluminally. Tesla-lineage; Bearden lineage; adopted wholesale by a large device literature.
- **The mainstream dialect.** There *is* a scalar sector in standard electrodynamics — the Coulomb-gauge scalar potential `φ`, which updates instantaneously over all space. This is a textbook fact, and the textbook verdict is that it is unobservable gauge artifact: every *field* stays retarded.
- **This framework's dialect.** Neither of the above. There is a specific degree of freedom, in a specific derived law, whose two settings give two different worlds — and we have measured the difference between them in-model rather than argued about it.

The third is the only one this paper defends. Its value is that it turns an unfalsifiable vibe into a number.

2 · Where it lives: inside the gravity law, not beside it

The framework's weak-field gravity result is a scalar deformation law relating the local time-rate factor to local energy density (The Shape of Gravity, which owns this result and its receipts):

> F = 1 + E / E★

DERIVED — with $\gamma_1 = 1$ and $E_\star = 2\ m_{edge}\ c^2$ in that paper's closure.

Now the observation this whole paper turns on. That law is dimensionally incomplete until you say *what* $E★$ *is*. It is a scale. A scale has to come from somewhere. And there are exactly two places it can come from:

reading	what $E_\star$ is	character of the law
fixed $E_\star$	a constant of nature, set once	strictly local; every influence inside a light cone
Machian $E_\star = \langle E\rangle(t)$	the instantaneous global mean energy	every clock's rate references the whole universe *now*

This is the entire content of "the scalar." Under the Machian reading, the normalization is a single global number that every local law consults, and it updates everywhere at once — not because anything travels, but because it is not a travelling thing. It is a *constraint*, in exactly the sense that Coulomb-gauge φ is a constraint.

> **The answer to "is the scalar connected to gravity?" is stronger than "connected."** In this framework the scalar field is not a separate field that couples to gravity. It *is* a degree of freedom of the gravity law — the one that fixes its scale. There is no scalar sector to go looking for somewhere else in the theory; if it is anywhere, it is here, and it has been here the whole time. DERIVED

The uncomfortable part, recorded honestly: **this project's own engine had silently chosen Machian, and this project's own gravity paper writes fixed.** Neither had noticed. The dichotomy was found by looking, not by reasoning — which is the correct order, and is why it produced a number instead of an argument.

3 · Both endpoints, measured

S4 (2026-07-19, pre-registration frozen before the run; receipts `lpoh/sims/PREREG_S4_2026-07-19.md`, `lpoh/sims/S4_two_channels.py`, `lpoh/sims/RESULTS_S4_2026-07-19.md`) perturbed a C_{a0} doctrine chamber and watched a far shell (distance ≥ 8) for pre-front deviation, under each reading with common random numbers.

experiment	result
wave channel, Machian `E⋆ = ⟨E⟩`	far-shell precursor **4.9×10^{-5} at t = 2**, against a causal wave arrival of 1.5×10^{-2} at t ≈ 8–16 — an instantaneous channel, ~300× below the wave amplitude
wave channel, fixed `E⋆`	front slope **0.91 ticks/edge**; far-shell deviation **exactly 0.00** until front arrival — a strict causal cone
consensus/repair layer (CRN-fixed)	**no fast channel at all**: perturbations heal in place (reach ≤ 2 edges in 40 sweeps); apparent speed 0.2 / 2.0 / 0.5 edges/sweep for random-sequential / Gauss–Seidel / Jacobi — i.e. *schedule gauge*, not physics
detonation vs chemical, equal energy, local law	identical fronts, no pre-front lead — frozen expectation confirmed

MEASURED (in-model). Three things worth extracting, because each kills a plausible alternative:

1. **The repair layer is not the scalar's home.** The obvious guess — that a "faster-than-light" sector would hide in the consensus/repair dynamics — is wrong in this stack, and measurably so. Repair *heals* rather than transmits, and what speed it appears to have is an artifact of the update schedule. This is a real negative and it narrows the search to one place.

2. **The precursor is weak but structured.** It carries the perturbation's timing everywhere at once. In-model, under Machian, it is usable-for-signalling *by construction*. That is a statement about the model, not about nature — see §8.
3. "**Nuclear is special" does not come from the wave channel.** At equal injected energy, detonation and chemical release produced identical fronts. If there is special content there it has to come from record depth (baryon-layer rewrites vs electron-shell rearrangement), and the sim has no record-depth hierarchy. Stated as a gap, not papered over.

4 · The dichotomy dissolves: the τ-family

S5 (same discipline, receipts `lpoh/sims/RESULTS_S5_2026-07-19.md`) asked the obvious follow-up: why should `E★` be either a frozen constant or an instantaneous mean? Make it a dynamical global scalar that *relaxes* toward `⟨E⟩` with timescale `τ`.

`τ`	**pre-front precursor (t = 2–4)**	
0 (Machian)	1.21×10^{-4}	S4's instant channel
1	1.21×10^{-4}	
4	5.75×10^{-5}	
16	2.17×10^{-5}	≈ 1/τ scaling
64	6.25×10^{-6}	
256	1.63×10^{-6}	
∞ (fixed)	**0 (exact)**	front slope 0.91 ticks/edge — the causal cone intact underneath

The precursor amplitude falls as `~1/τ` and the causal wave channel is **untouched at every `τ`**. MEASURED

> **This is the paper's second real result, and it reframes the whole subject.** Fixed and Machian are not rival theories. They are the two endpoints of one continuous family, and the

> physical question is not "is there a scalar?" — there is, structurally, always — but "**what is `τ_global`?**" A universe with `τ` of order the Hubble time is *Machian on cosmological timescales* (E★ tracks the evolving cosmic mean, which is what a Machian cosmology needs) and *indistinguishable from fixed-E★* in any laboratory or gravitational-wave-timescale experiment. Both readings become true, at different timescales. The either/or was an artifact of only ever looking at the endpoints.

One metric caveat, disclosed rather than buried: the threshold-based front-slope column in the S5 run saturates whenever any precursor exceeds 10^{-9} (it reads 0.08 for every finite `τ`). The amplitude column is the real observable; the `τ = ∞` row is what confirms the causal cone underneath.

5 · Why "scalar" is the right word, precisely

Three properties, each of which the name is doing real work to convey:

- **It is a scalar.** One number, not a vector or a tensor. There is no direction to it, and no polarization.
- **It is potential-like, not force-like.** It acts through the clock rate — through phase — not by pushing on anything. Nothing in the framework's force sector changes when you move along the τ-family.
- **It is a constraint, not a solution.** This is why "instantaneous" is not a speed claim and does not violate anything. A constraint does not propagate; it holds. Asking how fast Gauss's law travels is a category error, and asking how fast `E⋆` travels is the same category error.

That last point is the one most often lost. The fringe literature says "instantaneous propagation," which sounds like a velocity claim and invites an immediate relativity objection. The correct statement is that there is nothing propagating to assign a velocity to. The objection that actually bites is a different one, and it is §8's.

6 · Three scales, one structure

The reason to take the global sector seriously is not the gravity rung, which is open. It is that *the same structure — a local law plus one global scalar* — has already been shown to do real, measured, parameter-free work one scale down.

scale	local law	global scalar	status
atom	single-particle orbital equation	chemical potential / mean field	**demonstrated** (S5-E2, 6/6, control 0/6)
chamber	edge scatter + derived-F	`E★ ← ⟨E⟩` at rate 1/τ	**measured** (S4, S5-E1: precursor ~1/τ)
universe	patch repair	`T(𝕌) = 𝕌` · `E★`	OPEN (the no-signalling obligation, §8)

The atom rung is the strongest thing in this paper

Thomas–Fermi run as an OPH-style consensus — a local density law plus exactly one global scalar (the chemical potential), settled self-consistently, with **zero fitted constants** — reproduces the aufbau ordering pairs that define the periodic table's structure: 4s < 3d at K/Ca, 5s < 4d at Rb/Sr, 6s < 5d at Cs/Ba. **6/6.** The consensus-stripped bare-Coulomb control gets **0/6.** MEASURED

The control is what makes this a result rather than a curve fit: strip the global scalar and the ordering collapses entirely. So *screening = many-observer consensus* stops being a slogan. The global sector is not a speculative addition to the framework; at atomic scale it is the thing that makes the periodic table come out right, and it was already there.

What did *not* pass, recorded: the exponent gate G2 failed as frozen (0/6). The post-hoc diagnosis is itself clarifying — `s_eff` is 1.000 exactly at both competing density peaks (both sit in the Latter tail), while the screening zone's local exponent runs 1.5 → 2.7. So the exact-Bessel `s = 3/2` is an *aggregate over the competition region*, not a pointwise slope. The constant stays underived, now with a sharper statement of why. OPEN

7 · The same gap twice — a closure route, not a closure

This section exists because a reader who has just come from Magnetism will notice something, and it is better stated carefully here than left to be noticed carelessly.

Magnetism's honest boundary is Corollary 6.6d of the external OPH programme: the consistency requirements force the gauge algebra, force the connection, and force `F =`

`dA` — but they do *not* supply the Maxwell action or its kinetic coefficient. In the source's own words, the reconstruction "*do[es] not themselves supply this action or its nonzero kinetic coefficient*." What is missing is a single global scale on a quadratic form: the `1/g²` in `S = −1/(2g²) ∫ F ∧ *F`.

Set the three open items of this programme side by side:

object	**what is forced**	**what is not**
Maxwell action (OB-2 / Cor. 6.6d)	the quadratic form `F ∧ *F`	the coupling `1/g²` — one global scale
gravity law (§2)	`F = 1 + E/E⋆`	the scale `E⋆` — one global normalization
repair generator (`canonical_seam_repair.py`)	uniquely `−L`, forced band ratio `(5−√5 : 6 : 5+√5)`	the time-scale — one residual global scale

Three statements of the same shape: *forced form, unfixed global normalization.* And there is already a real, receipted narrowing along this line. Mueller's own instrument (`oph_fpe/gauge/kinetic_selector_sweep.py`) asked whether positivity, locality on the twelve-port carrier, and A5 covariance select a unique quadratic response law; run directly, **they do not** — twelve candidates survive, with a constructive two-law counterexample and receipts `UNIQUE_GAUGE_KINETIC_SOURCE_RAY_RECEIPT: false`. But checking those twelve against `canonical_seam_repair.py`'s forced band ratio leaves **exactly two** (`edge_repair`, H = I + L; `double_edge_repair`, H = I + 2L) — and those two are related by precisely the scale freedom the repair generator already allows. Twelve candidates collapse to a one-parameter family.

> **The conjecture, stated so it can be killed.** If the residual time-scale freedom of the repair generator, the scale `E⋆`, and the Maxwell coupling `1/g²` are *the same* global normalization degree of freedom, then **fixing `τ_global` fixes the electromagnetic coupling**, and the two open problems are one open problem with one answer.
>
> **What is wrong with this as it stands.** (i) The bridge premise — that the response Hessian should match the repair generator — is this project's, not an OPH theorem, and is unproven. (ii) The 12 → 2 narrowing is composed from two already-run certificates and introduces no new computation; it is real but it is arithmetic on other people's results. (iii) Even granted

> entirely, it is *silent* on the radiative-vs-non-radiative question that is OB-2's actual content. (iv) Most importantly, the external synthesis's own Definition 1.2 — *resemblance is insufficient* — is the standard this project applies to everyone else, and it applies here: "same shape" is not "same object," and nothing above supplies the argument that would upgrade it. OPEN

So this section claims exactly one thing: **a route**. The Maxwell-action gap is not derived here and must not be cited as derived here — OB-2 remains its owner and remains open. What this paper adds is the observation that the gap has the same shape as the one the τ-family gives a *measured handle on*, and that this makes `τ_global` a more valuable number than it looked when it was only about gravity.

8 · Anchors, and the thing that actually threatens the idea

What is measured in mainstream physics

- **Coulomb gauge** — the scalar potential updates instantaneously over all space. Textbook. The standard verdict is "unobservable gauge artifact"; the entire fringe claim is that this sector is physical. The translation into this framework is exact: *gauge freedom = freedom in the record normalization*, and "the scalar is physical" = "the normalization is Machian and couples." MEASURED (the gauge structure) / OPEN (its physicality)
- **Aharonov–Bohm** — potentials have observable consequences where fields vanish. The standing precedent that "potential-like" does not mean "fictitious." MEASURED
- **GW170817** — the speed of gravity equals c to one part in 10^{15}. This bounds the *wave* channel and nothing else. A weak normalization precursor in a different channel is neither excluded nor supported by it. MEASURED
- **Bell tests, loophole-free (2015, three independent labs)** — nature has instantaneous *correlations* that no subluminal account produces. This is the one place the global sector is already measured in a laboratory. The framework's reading: entangled systems *share a record*, i.e. are adjacent in the record layer regardless of metric distance. MEASURED
- **Mach's principle** — the `E★ = ⟨E⟩` reading is literally Machian inertia. General relativity partially incorporates and partially rejects Mach; the question is old, respectable, and unsettled.

The threat

Not relativity — §5 disposes of the naive velocity objection. The threat is the **no-communication theorem**. Bell correlations are instantaneous and provably cannot carry a controllable signal. Nature, where we have looked, has consistently arranged for the global sector to be real and unusable at the same time.

> **The owed theorem, stated as an obligation.** This framework has **no repair-layer no-signalling theorem**. Until one exists — proving or refuting that boundary-conditioned repair can be modulated to carry controllable signal — **the framework neither licenses nor forbids a usable instantaneous channel**, and every claim in either direction is a modelling choice wearing a physics costume. In-model usability under Machian normalization (§3) is a fact about our engine, not a permission slip. This is logged as a programme-level obligation and is the single highest-value thing anyone could prove here. OPEN

What else is owed

- **The record-depth hierarchy.** The sim has no baryon-vs-shell depth structure, so the "nuclear is loud in the record layer" mapping is not reproduced and is not asserted. OPEN
- **The parked falsifiable prediction.** If an instrument operating on the normalization sector ever exists, disruption amplitude should scale with **transmuted mass** (records rewritten), not with **yield** (energy released) — so equal-yield fission and fusion devices should be distinguishable. Parked until there is an instrument; it is the mapping's falsifiable content and it should not be quietly dropped. OPEN
- **The universal discriminator.** Any Machian or preferred-frame coupling inherits one clean signature everywhere: modulation at the **sidereal period, 23 h 56 m**, where no mundane artifact lives. Any bench claiming to touch this sector should be analysed at sidereal period first and everything else second.

9 · What this is not

> **Firewall.** Five confusions this paper should prevent rather than create.
>
> 1. **Not a fifth force.** It exerts nothing. It normalizes.

2. **Not the "scalar" of scalar magnetometry.** A *scalar magnetometer* measures |B|, the rotation-invariant magnitude of the magnetic field, as opposed to its vector components. That is an ordinary, well-understood, entirely mundane instrument, and the word collision with this paper is pure coincidence. There is **no derived coupling** from the global sector to any magnetometer, and nothing here should ever be offered as a detection mechanism for one. See [Magnetism] for what a magnetometer actually measures.
3. **Not a licensed communication channel.** See §8. The theorem that would settle it does not exist.
4. **Not support for over-unity or anti-gravity.** The scalar-EM device literature's core intuition — that there is a scalar, quadratic, global sector standard transverse bookkeeping drops — lands structurally near this result, and that is worth recording. Its device claims do not follow from it and are not supported by it. A squared sine is time-symmetric; phase conjugation is not anti-gravity.
5. **Not a derivation of the Maxwell action.** §7 is a route with four stated defects. Cite OB-2, not this paper.

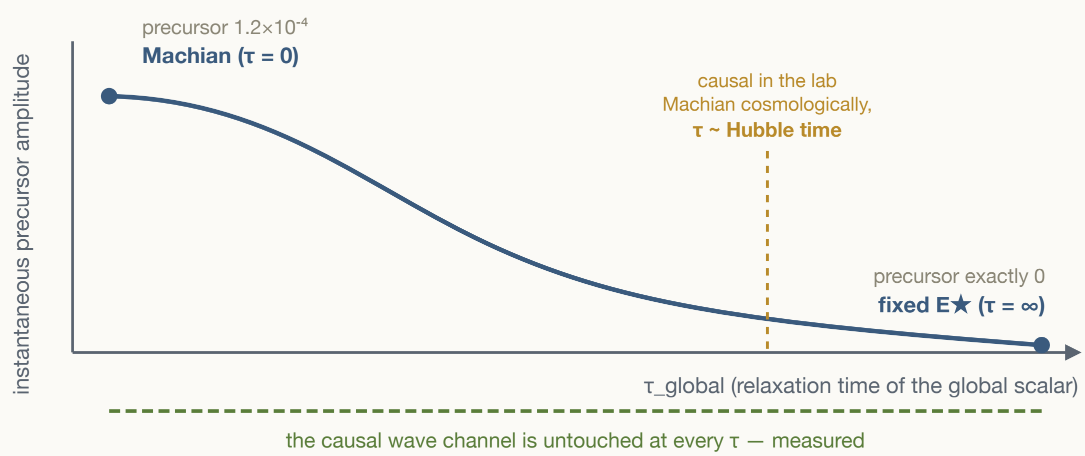

Figure 1 — the τ-family. The "is there an instantaneous scalar channel?" question is not binary. Fixed and Machian `E⋆` are the two endpoints of one continuous family indexed by how fast the global scalar relaxes, and the instantaneous precursor's amplitude falls off as `1/τ` between them. The causal wave channel — the one GW170817 pins to *c* — is unaffected everywhere along the axis. A universe sitting near the amber line is Machian on cosmological timescales and indistinguishable from strictly local in any laboratory.

Claim boundary. This paper introduces no new simulation and no new theorem. Every measured number is S4's or S5's, pre-registered before the run, and is cited to `lpoh/sims/RESULTS_S4_2026-07-19.md` and `lpoh/sims/RESULTS_S5_2026-07-19.md`. The gravity law `F = 1 + E/E⋆` belongs to The Shape of Gravity — cite that paper, not this one, for it. The gauge-forcing theorem, the curvature identity, and the holonomy theorem belong to the external OPH papers cited in Magnetism's sources box — cite those. §7 is a route, not a result: **do not cite this paper as deriving the Maxwell action or its coupling.** All S4/S5 numbers are *in-model* measurements — they establish what this framework's engine does under each reading, and establish nothing whatsoever about which reading nature implements. The Greer-testimony and Steiner mappings that motivated the original S4 investigation are deliberately absent from this paper's body; they live in `papers/SCALAR_CHANNEL_OPH_ANALYSIS_2026-07-19.md` §§4–5, tagged there as working-hypothesis and postdictive-at-best, and nothing in this paper depends on them.

Sources and receipts.

- Pre-registrations, frozen before their runs: `lpoh/sims/PREREG_S4_2026-07-19.md`, `lpoh/sims/PREREG_S5_2026-07-19.md`.
- Code and logs: `lpoh/sims/S4_two_channels.py`, `lpoh/sims/S5_global_sector.py`, `lpoh/sims/S4_run.log`, `lpoh/sims/S5_run.log`.
- Results: `lpoh/sims/RESULTS_S4_2026-07-19.md`, `lpoh/sims/RESULTS_S5_2026-07-19.md`. Both runs disclose a harness artifact caught and fixed mid-run (S4: twin RNG-stream desync, fixed with common random numbers; S5: a max/min sign inversion caught by its own giveaway).
- Working analysis, including the motivating testimony mapping this paper firewalls out: `papers/SCALAR_CHANNEL_OPH_ANALYSIS_2026-07-19.md`.
- The Machian option's classical lineage (Ampère → Weber → Assis) and the open-hairpin discriminator: `papers/AMPERE_LONGITUDINAL_SECTOR_NOTE_2026-07-20.md`.
- §7's two certificates, both in `muellerberndt/oph-physics-sim`: `oph_fpe/gauge/kinetic_selector_sweep.py` and `oph_fpe/dynamics/canonical_seam_repair.py`. The 12 → 2 composition is recorded in OB-2's third note.